

Conditional Statements in Python

1. if Statement

Used to execute a block of code when a condition is True.

```
age = 20

if age >= 18:
    print("You are an adult.")
```

2. if-else Statement

``else`` executes when the ``if`` condition is False.

```
age = 16

if age >= 18:
    print("Eligible to vote.")
else:
    print("Not eligible to vote.")
```

3. if-elif-else

Used when there are multiple conditions. Only the first true condition is executed.

```
marks = 75

if marks >= 90:
    print("A Grade")
elif marks >= 60:
    print("B Grade")
else:
    print("C Grade")
```

4. Comparison Operators

`==` Equal to `!=` Not equal to `<` Less than `>` Greater than `<=` Less than or equal to `>=` Greater than or equal to

```
age = 19

if age >= 18:
    print("Eligible")
```

5. Logical Operators

``and`` → both conditions must be true. ``or`` → at least one condition must be true. ``not`` → reverses the condition.

```
age = 16
has_id = True

if age < 18 and has_id:
    print("Eligible for discount")
```

6. Nested if

An ``if`` statement inside another ``if`` statement is called a nested if.

```
day = "Sunday"
raining = False
```

```
if day == "Sunday":
    if not raining:
        print("Let's go outside!")
```

7. Indentation

Python uses indentation to define code blocks. Usually, 4 spaces are used.

```
if age >= 18:
    print("Eligible")
    print("Bring your ID.")
else:
    print("Not eligible")
```

8. match-case (Python 3.10+)

Useful when comparing one value against several possible values. `\_` acts like the default case.

```
day = "Sunday"

match day:
    case "Monday":
        print("Work day")
    case "Saturday" | "Sunday":
        print("Weekend")
    case _:
        print("Other day")
```

Quick Difference

Statement	Purpose
if	Checks one condition
elif	Checks additional conditions
else	Executes when all conditions are false
Nested if	if inside another if
match-case	Matches one value against multiple patterns

Important: Python relies on indentation, so incorrect indentation can cause an error or change the meaning of the program.